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RALPH O'CONNOR SUSTAINABLE ENERGY INSTITUTE

Market Signals and Incentives Aren't Sufficient: The Case for Supply-Chain Modeling for Grid Expansion

Boyu Yao, Yury Dvorkin
10/27/2025

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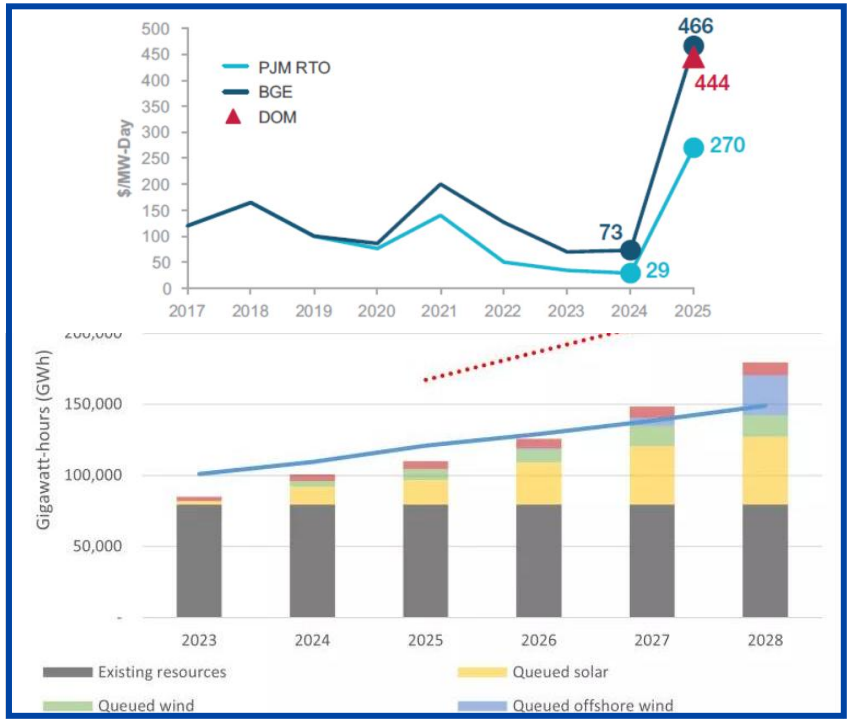
Energy at Hopkins

Two stories reveal the mismatch between plans and supply chains

- Story 1 — New York Wind: The Broken Promise
  - **2021:** Contracts for major offshore wind farms (e.g., Empire Wind 2) promised power for millions [1].
  - Contracts locked in, market signals clear, OREC incentives secured [2], developers ready to build.
  - **By 2024:** Inflation, **turbine and cable shortages**, and supply chain bottlenecks halted projects [3].
  - Contracts collapsed — markets pointed forward, incentives aligned, but **supply chains held back**.

- Story 2 — PJM Market: The Price Shock
  - **2024:** PJM capacity auction prices **spiked up to 800%** in some zones [4].
  - Consumers and businesses faced billions in extra costs**.**
  - Projects stalled in queues, **transformers, materials**, and financing all **in short supply** [5].
  - Market signals were strong, but **physical bottlenecks** blocked expansion.



[1] https://www.nyserda.ny.gov/All-Programs/Offshore-Wind/Focus-Areas/NY-Offshore-Wind-Projects
[2] https://www.nyserda.ny.gov/All-Programs/Offshore-Wind/Focus-Areas/Offshore-Wind-Solicitations/2020-Solicitation
[3] https://patch.com/new-jersey/longbranch/empire-wind-2-turbines-canceled-would-have-been-19-miles-nj
[4] https://www.reuters.com/business/energy/pjm-power-auction-results-yield-sharply-higher-prices-2024-07-31
[5] https://opc-dc.gov/wp-content/uploads/2025/05/PJM-Capacity-Market-Report-FINAL-OPC-Synapse.pdf

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Incentives ≠ Deployment!

Supply chain, not contracts, dictated what was feasible.

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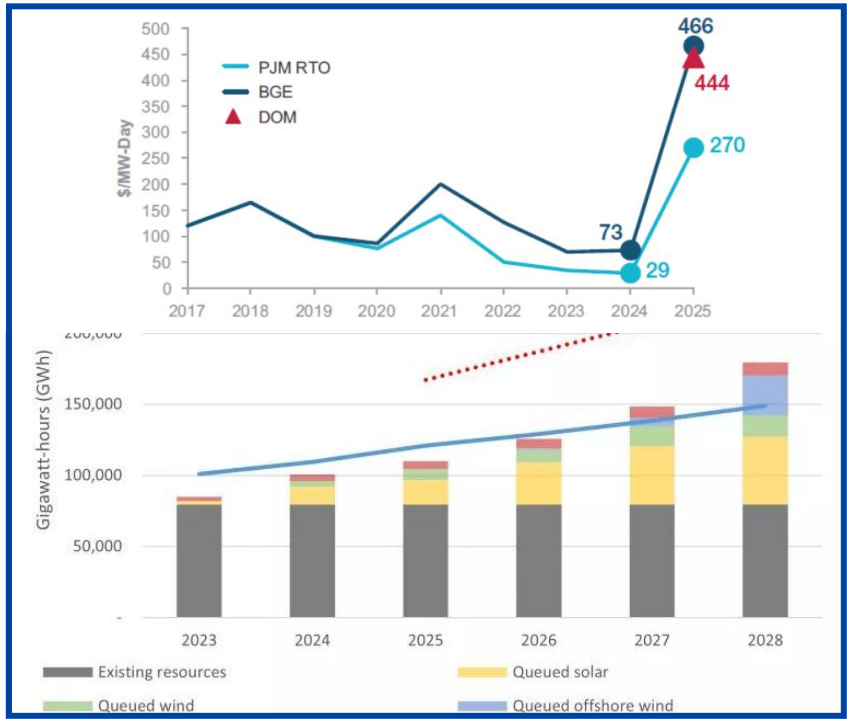

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JANUARY 5, 2024

Offshore wind contracts with Empire Wind 2 cancelled

by WorkBoat Staff in Wind, Offshore, News

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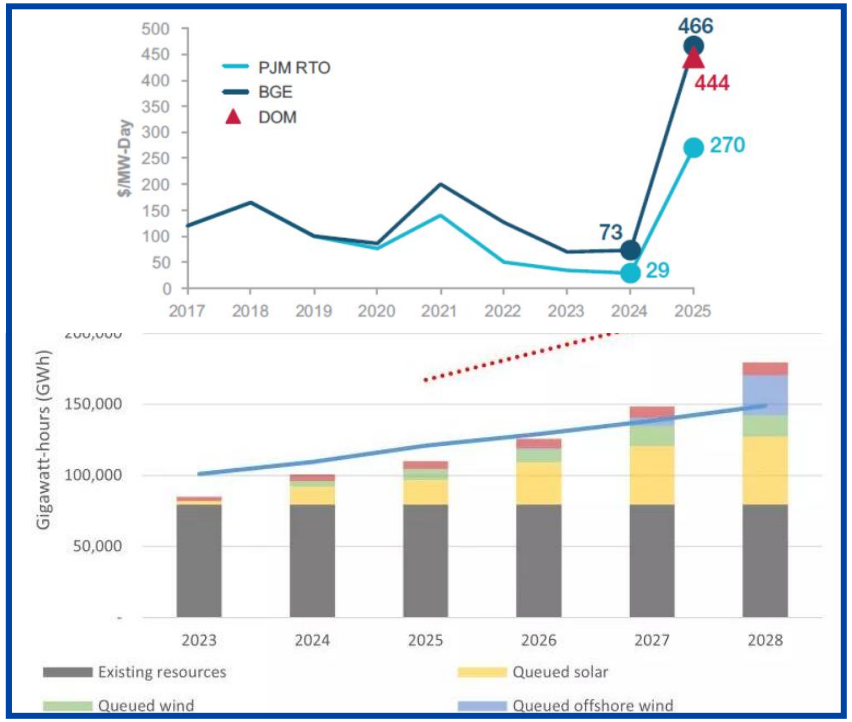
Prices ≠ Projects!

Supply chain, not market signals, determined what could proceed.



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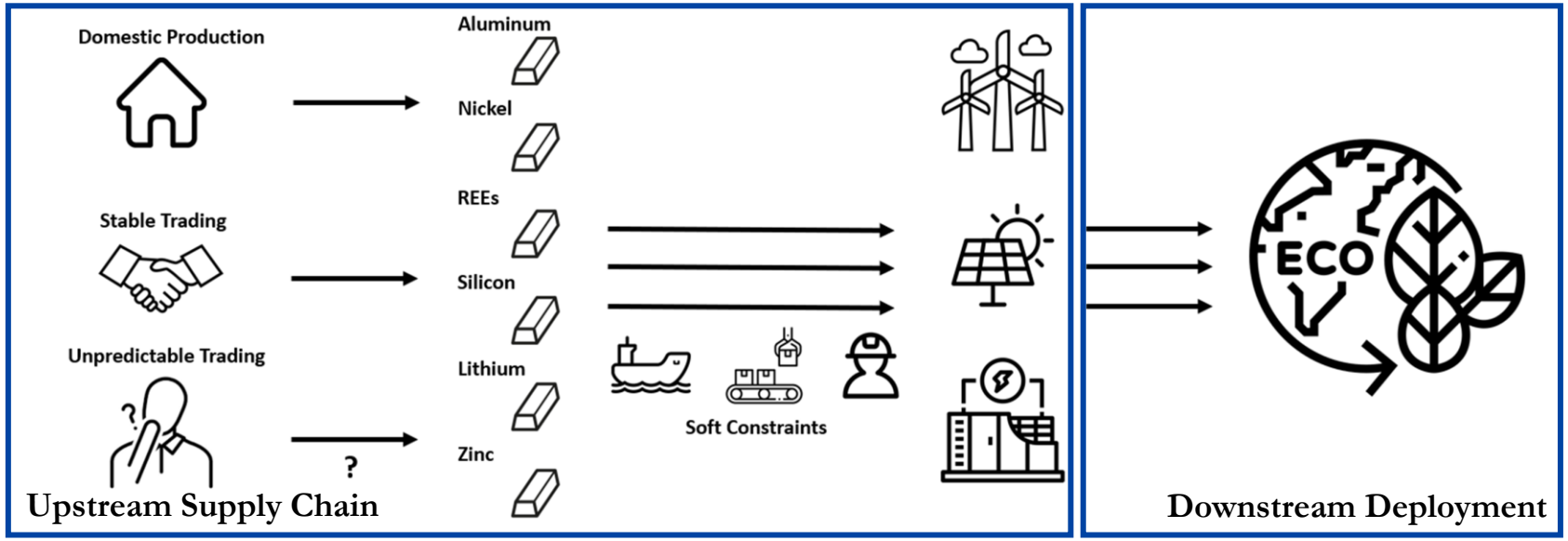
Gap 1: Insufficient visibility into upstream supply chains



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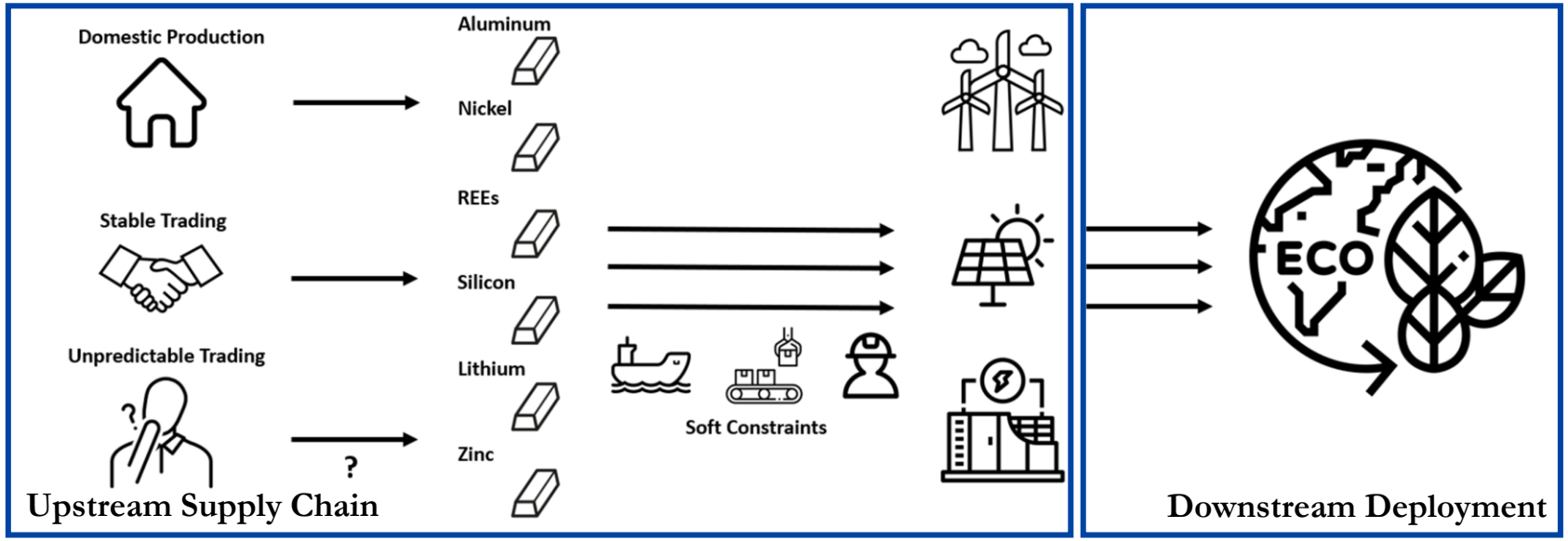
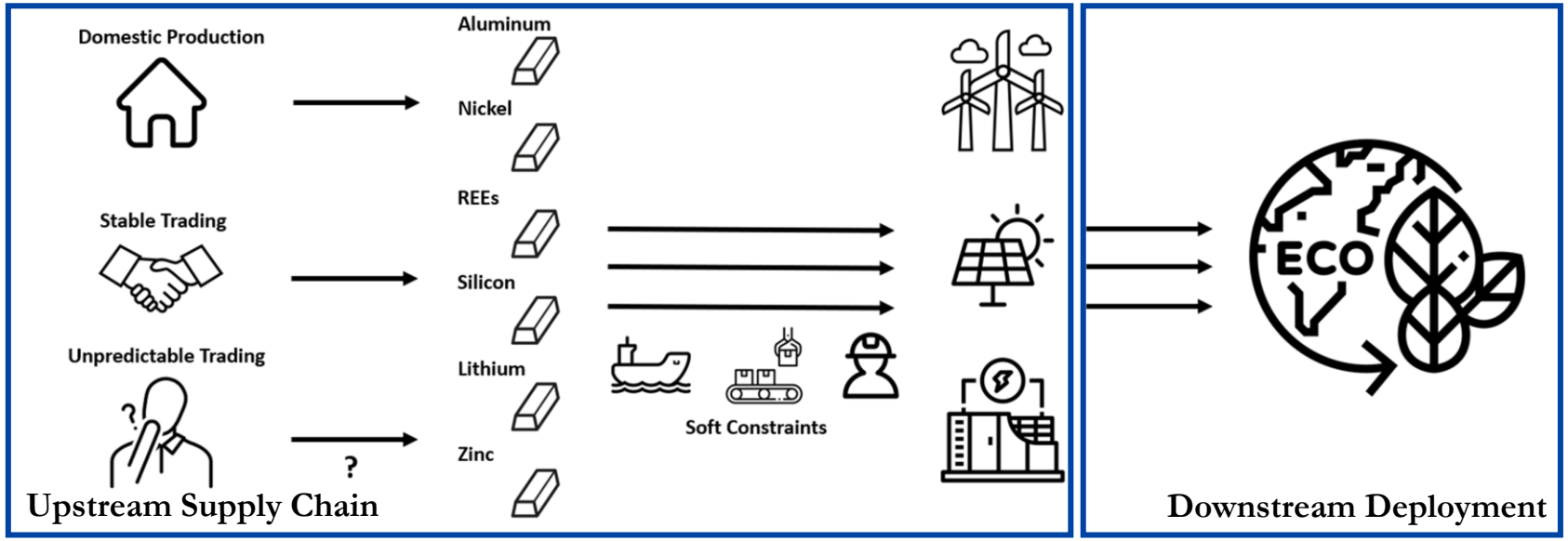


Materials → Components → Products

Upstream data (materials, lead times, bottlenecks) **is scarce**.

Downstream data (load, generation, prices) **is rich**.

Supply chain constraints for the US clean energy transition [8, 9]

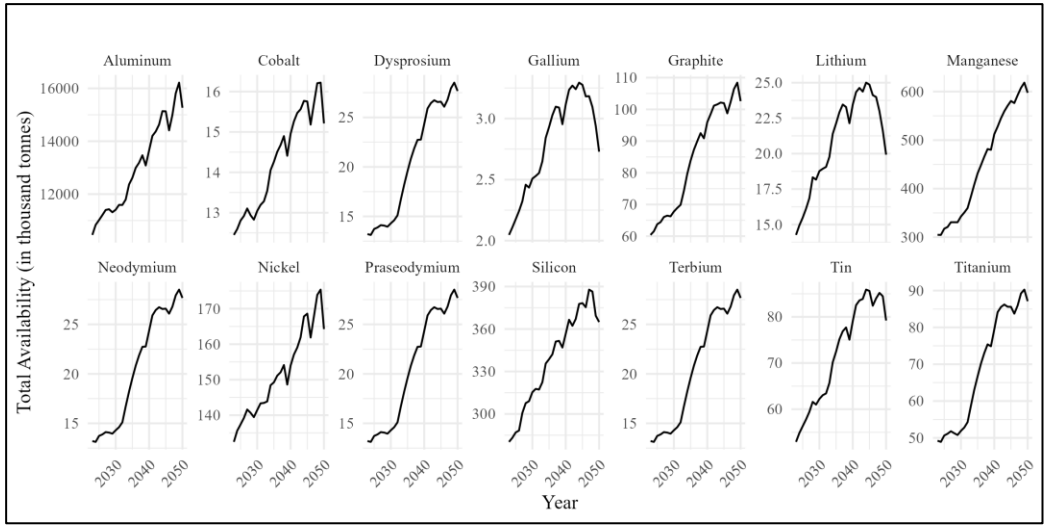


Materials → **Components** → **Products**

26 Critical materials from USGS and DOE critical lists [10, 11]; Sources categorized as: Stable vs. Unpredictable

Projecting Future Availability →

***Supercycle:** Long-term price and demand trends driven from global technology and resource use.



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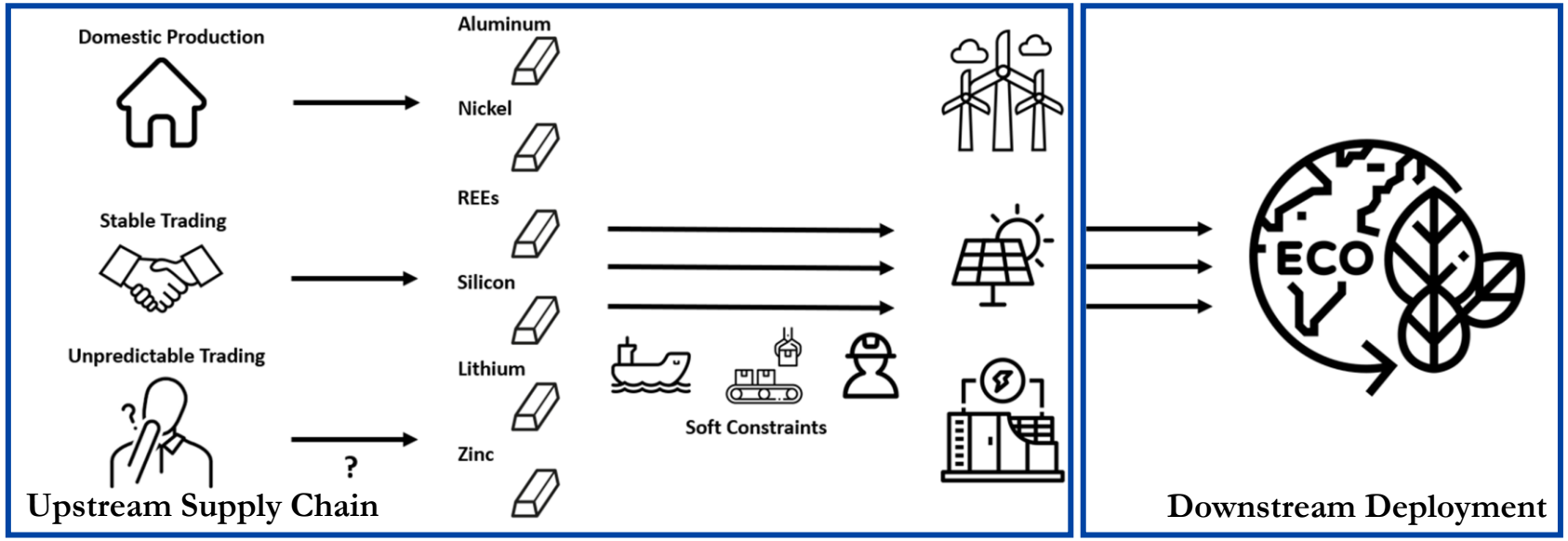
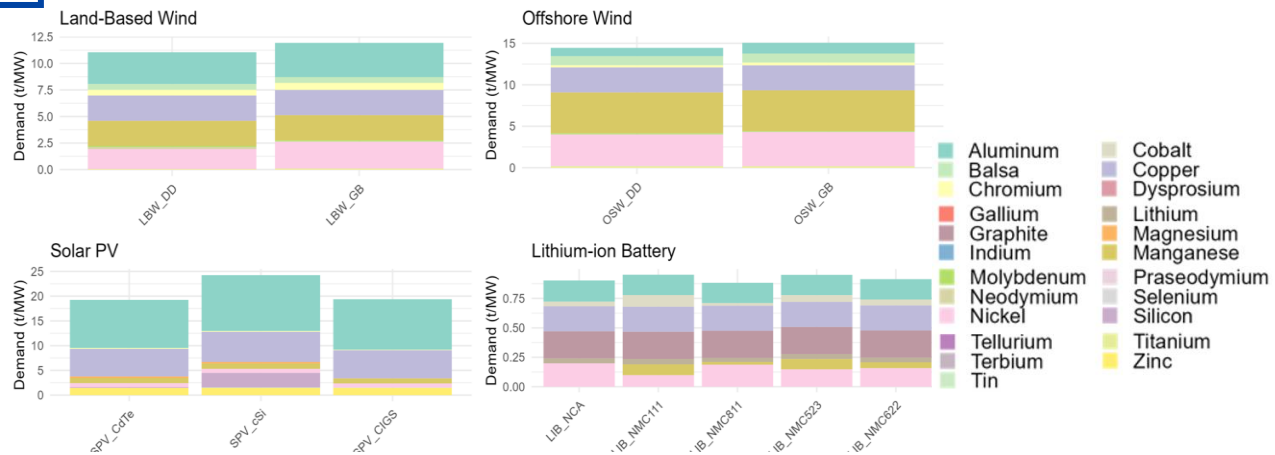


Materials → Components → Products

Established **our** dataset on material intensity (e.g., tons/MW).

Wind: Onshore & Offshore (Gearbox, Direct-drive)
Solar PV: c-Si, CIGS, CdTe
Batteries: Li-ion (NCA, NMC-111/523/622/811, LFP)



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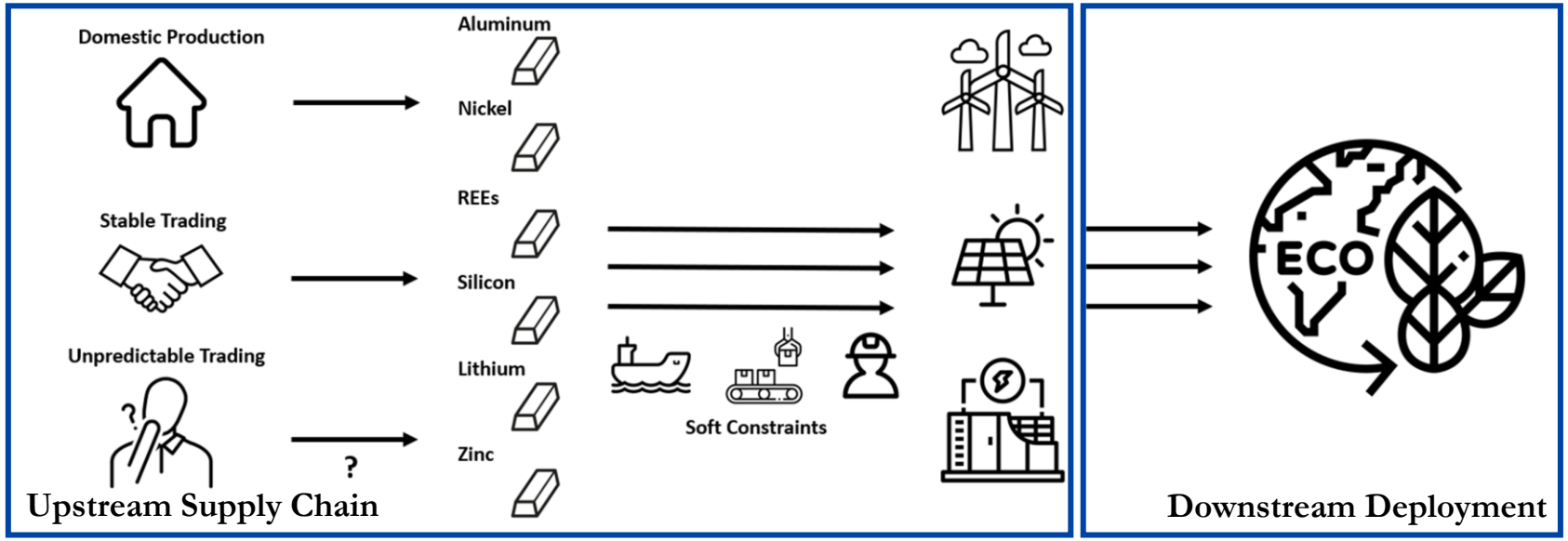
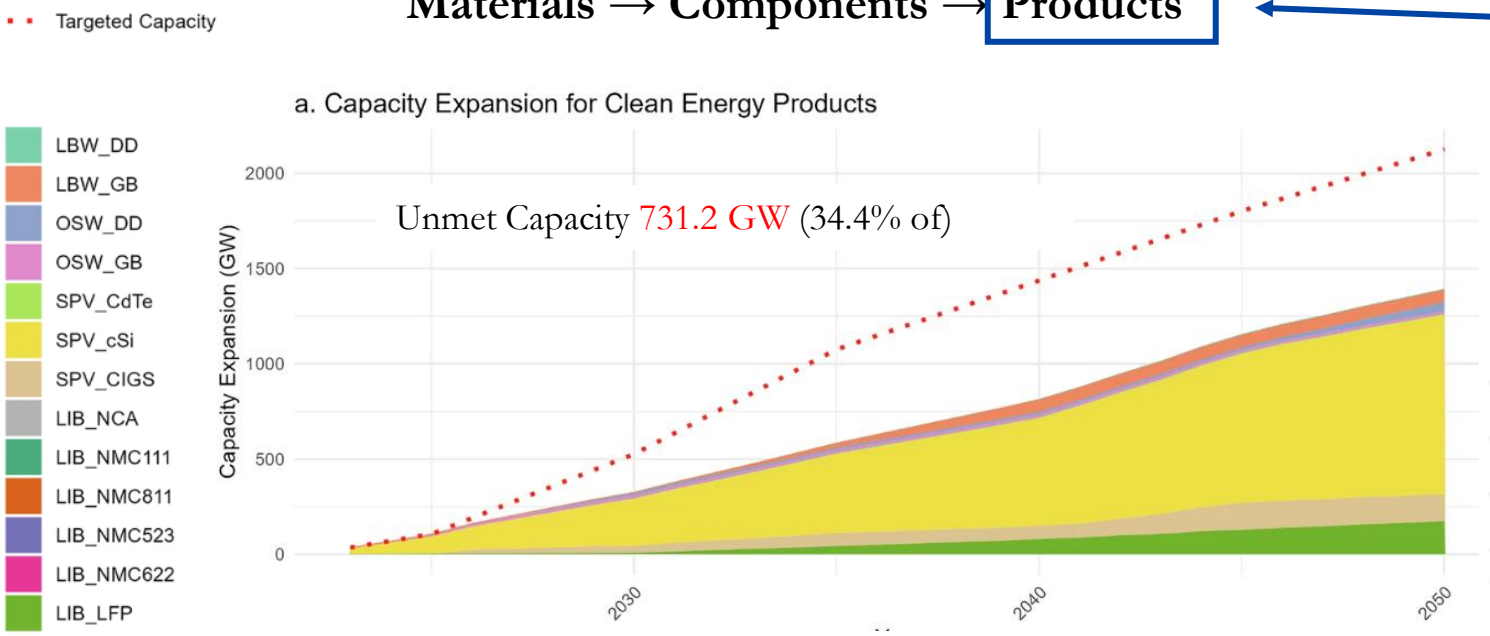


Materials → Components → Products

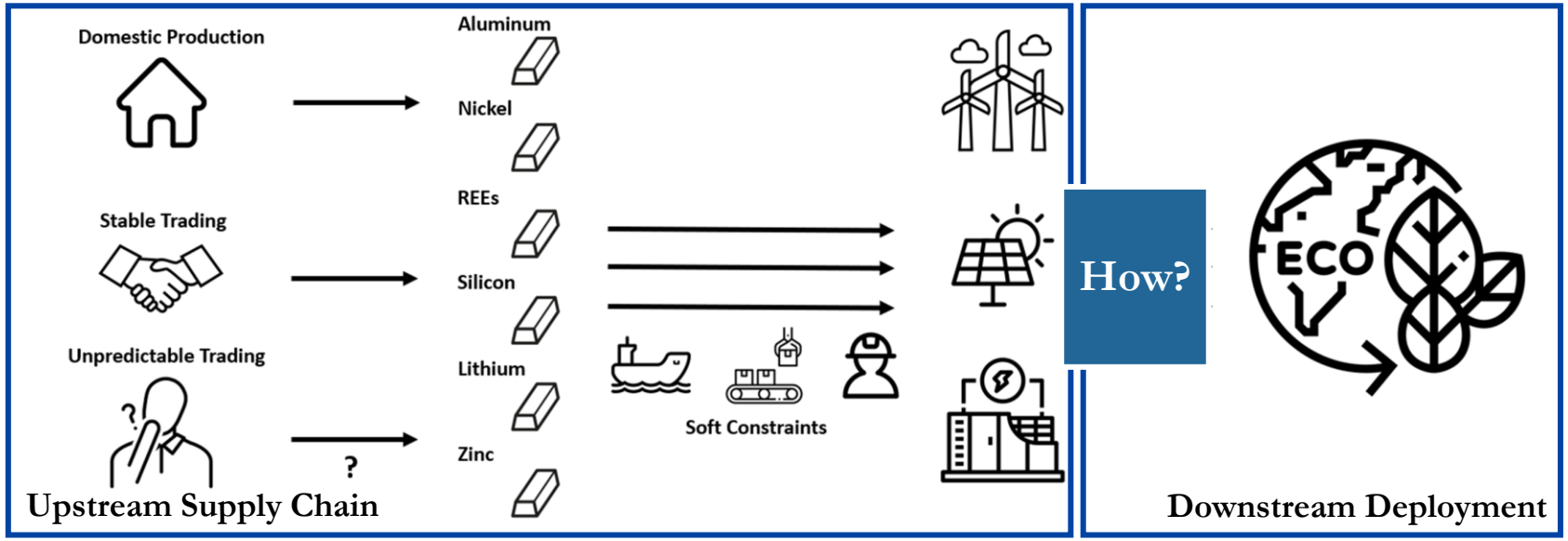
2050 Net-Zero:
Land-based wind: 400 GW;
Offshore wind: 110 GW;
Solar PV: 1,600 GW;
Battery storage: 125 GW.



Extra materials needed:

- **Neodymium:** 7,591 metric tons (+67.5%)
- **Nickel:** 1,355,659 metric tons (+106.2%)
- **Silicon:** 913,258 metric tons (+32.1%)
- **Tellurium:** 2,834 metric tons (+327%)

Gap 2: No unified model linking supply chains to capacity expansion planning



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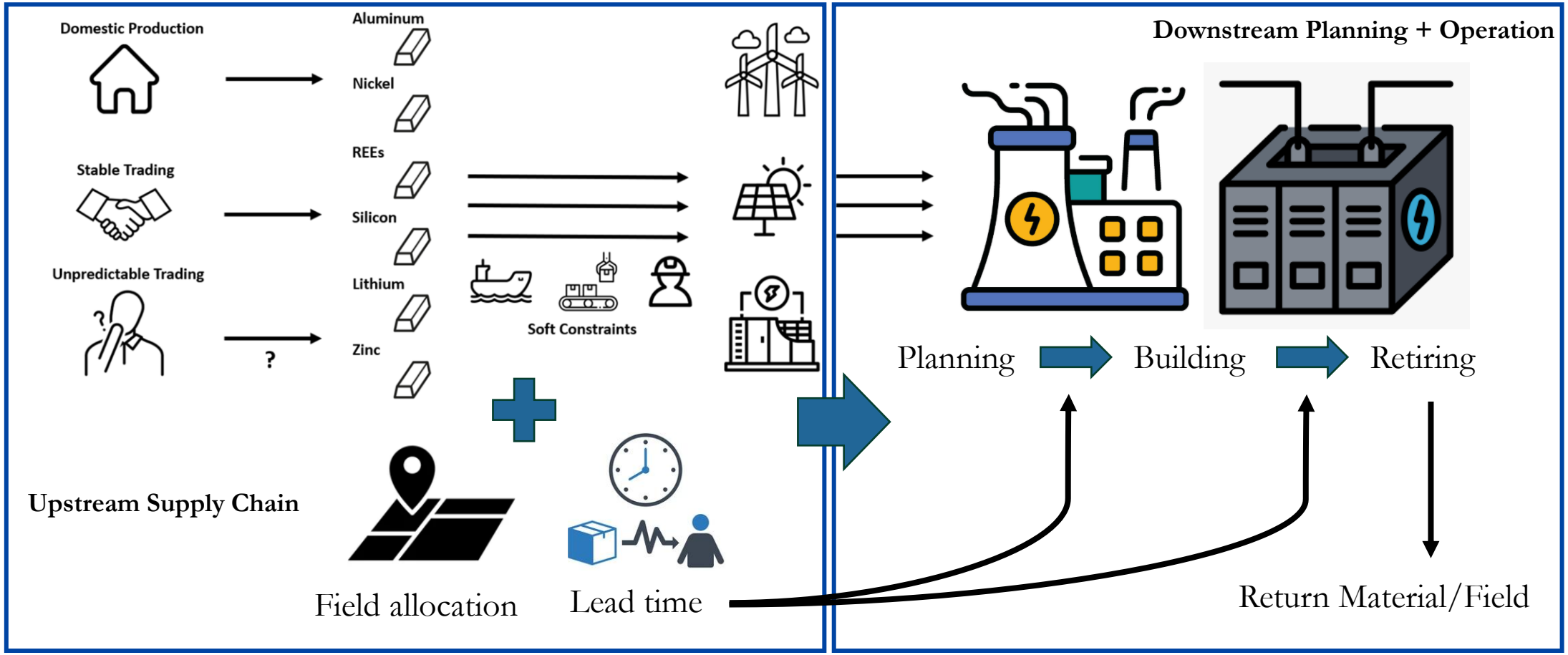

Integrating upstream supply chains into generation expansion planning [12]

obj.

$$\min \sum_y C_y = \sum_y \left(C_y^{\text{in}} + C_y^{\text{op}} + C_y^{\text{pe}}\right) \quad (1a)$$

$$C_y^{\text{in}} = \sum_{g \in \bar{\mathcal{G}}} AC_{gy}^{\text{I}} \cdot \overline{P}_g^{\text{G}} \cdot d_{gy} \quad (1b)$$

$$C_y^{\text{op}} = \sum_g C_g^{\text{F}} \cdot \overline{P}_g^{\text{G}} \cdot o_{gy} + \sum_{g \in \mathcal{G}^{\text{th}} \cup \mathcal{G}^{\text{rn}}} \sum_t C_g^{\text{V}} \cdot N_{ty} \cdot \sum_h p_{gthy} + \sum_{g \in \mathcal{G}^{\text{st}}} \sum_t C_g^{\text{V}} \cdot N_{ty} \cdot \sum_h (c_{gthy} + dc_{gthy}) \quad (1c)$$

$$C_y^{\text{pe}} = \sum_t N_{ty} \cdot P^{\text{VOLL}} \cdot \sum_h p_{ithy}^{\text{LS}} + P^{\text{RM}} \cdot p_y^{\text{RM}} + P^{\text{RPS}} \cdot e_{ky}^{\text{RPS}} \quad (1d)$$

1.

$$u_{my} \geq \sum_c v_{cy} \cdot D_{mc}^{\text{CO}}, \quad \forall m, y \quad (2a)$$

$$v_{cy} \geq \sum_p w_{py} \cdot D_{cp}^{\text{PR}}, \quad \forall c, y \quad (2b)$$

$$u_{my} \leq \overline{M}_{my} + \sum_{g \in \mathcal{G}} R_{mg}^{\text{RM}} \cdot \overline{P}_g^{\text{G}} \cdot r_{gy} + s_{my}, \quad \forall m, y \quad (2c)$$

$$s_{my} = s_{m(y-1)} + \overline{M}_{m(y-1)} - u_{m(y-1)} + \sum_{g \in \mathcal{G}} R_{mg}^{\text{RM}} \cdot \overline{P}_g^{\text{G}} \cdot r_{g(y-1)}, \quad \forall m, y \quad (2d)$$

2.

$$\sum_{g \in \mathcal{G}^k} \overline{P}_g^{\text{G}} \cdot d_{gy} \leq \sum_{p \in \mathcal{P}^k} w_{py}, \quad \forall k, y \quad (2e)$$

$$\frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot d_{gy}}{R_k^{\text{CAP}}} \leq f_{iy}^k, \quad \forall i, y, k \quad (2f)$$

3.

$$f_{iy}^k = f_{i(y-1)}^k + \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot r_{gy}}{R_k^{\text{CAP}}} - \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot d_{g(y-1)}}{R_k^{\text{CAP}}}, \quad \forall i, y > 1, k \quad (2g)$$

$$f_{i1}^k = A_i^k + \frac{\sum_{g \in \mathcal{G}^k \cap \mathcal{G}_i} \overline{P}_g^{\text{G}} \cdot r_{g1}}{R_k^{\text{CAP}}}, \quad \forall i, k \quad (2h)$$

4.

$$\sum_{y'' \leq y - T_g^{\text{LEAD}}} d_{gy''} = \sum_{y' \leq y} b_{gy'}, \quad \forall g \in \bar{\mathcal{G}}, \ y \quad (2i)$$

$$b_{gy} = 0, \quad \forall g \in \bar{\mathcal{G}}, \ y \quad (2j)$$

$$\sum_{y'' \leq y - T_g^{\text{LT}}} b_{gy''} = \sum_{y' \leq y} r_{gy'}, \quad \forall g \in \bar{\mathcal{G}}, \ y \quad (2k)$$

$$r_{gy} = 1 \text{ if } y = T_g^{\text{RT}}, \ 0 \text{ if } y < T_g^{\text{RT}}, \quad \forall g \in \bar{\mathcal{G}}, \ \forall y \quad (2l)$$

Objective =
Investment + Operations + Penalties

Upstream supply chain constraints

1. Materials → components → products.
2. Field availability.
3. Recovery from retired units.
4. Limits by production & lead times.

5.

$$\sum_{g \in \mathcal{G}_i \cap (\mathcal{G}^{\text{th}} \cup \mathcal{G}^{\text{rn}})} p_{gthy} + \sum_{g \in \mathcal{G}_i \cap \mathcal{G}^{\text{st}}} (dc_{gthy} - c_{gthy}) - \sum_{l \in \mathcal{LS}_i} q_{lthy} + \sum_{l \in \mathcal{LR}_i} q_{lthy} = L_{ithy} - p_{ithy}^{\text{LS}}, \quad \forall i, t, h, y \quad (3a)$$

6.

$$0 \leq p_{gthy} \leq \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{th}}, \ y, t, h \quad (3b)$$

$$0 \leq p_{gthy} \leq F_{igthy}^{\text{GEN}} \cdot \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}_i \cap \mathcal{G}^{\text{rn}}, \ i, y, t, h \quad (3c)$$

7.

$$o_{gy} = o_{g(y-1)} + b_{gy} - r_{gy}, \quad \forall g \in \mathcal{G}, \ y \geq 1 \quad (3d)$$

$$o_{g0} = 1 \text{ if } g \in \bar{\mathcal{G}}, \ 0 \text{ otherwise}, \quad \forall g \in \mathcal{G} \quad (3e)$$

$$-\overline{P}_l^{\text{L}} \leq q_{lthy} \leq \overline{P}_l^{\text{L}}, \quad \forall l, y, t, h \quad (3f)$$

$$\sum_{g \in \mathcal{G}^k} \overline{P}_g^{\text{G}} \cdot F_{ky}^{\text{ELCC}} \cdot o_{gy} + p_y^{\text{RM}} \geq (1 + R_y^{\text{RM}}) \cdot \overline{L}_y, \quad \forall y, k \quad (3g)$$

$$0 \leq p_{ithy}^{\text{LS}} \leq L_{ithy}, \quad \forall i, y, t, h \quad (3h)$$

8.

$$\sum_{g \in \mathcal{G}^k} \sum_t \sum_h N_{ty} \cdot p_{gthy} + e_{ky}^{\text{RPS}} \geq R_{ky}^{\text{RPS}} \cdot \sum_t \sum_h \sum_i N_{ty} \cdot L_{ithy}, \quad \forall k, y \quad (3i)$$

9.

$$0 \leq c_{gthy} \leq \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{st}}, \ y, t, h \quad (4a)$$

$$0 \leq dc_{gthy} \leq \overline{P}_g^{\text{G}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{st}}, \ y, t, h \quad (4b)$$

$$0 \leq e_{gthy}^{\text{soc}} \leq \overline{E}_g^{\text{ST}}, \quad \forall g \in \mathcal{G}^{\text{st}}, \ y, t, h \quad (4c)$$

$$e_{gthy}^{\text{soc}} = e_{gt(h-1)y}^{\text{soc}} + \epsilon^{\text{CH}} \cdot c_{gthy} - \frac{dc_{gthy}}{\epsilon^{\text{DC}}}, \quad \forall g \in \mathcal{G}^{\text{st}}, \ y, t, h \quad (4d)$$

$$e_{gt1y}^{\text{soc}} = e_{gtendy}^{\text{soc}} = 0.5 \cdot \overline{E}_g^{\text{ST}} \cdot o_{gy}, \quad \forall g \in \mathcal{G}^{\text{st}}, \ y, t \quad (4e)$$

Downstream operation constraints

5. Power Balance.
6. Generation, line, storage limits.
7. Evolution: unit builds & retirements (binary).
8. Reserve margin (ELCC) + policy compliance.
9. Battery charge/discharge, SOC.

[12] Yao, B., Bernstein, A., & Dvorkin, Y. (2025). Integrating Upstream Supply Chains into Generation Expansion Planning. arXiv preprint arXiv:2508.03001.

Integrating upstream supply chains into generation expansion planning [12]

- **Decisions (annual): where/when/what** to build; **product mix** (Solar/Wind/Battery variants).
- **Operations (hourly, rep. days):** dispatch, storage charge/discharge, flows, load shedding.
- **Upstream layers:** materials → components → products; lead time; stocks & recovery from retirements; field availability.

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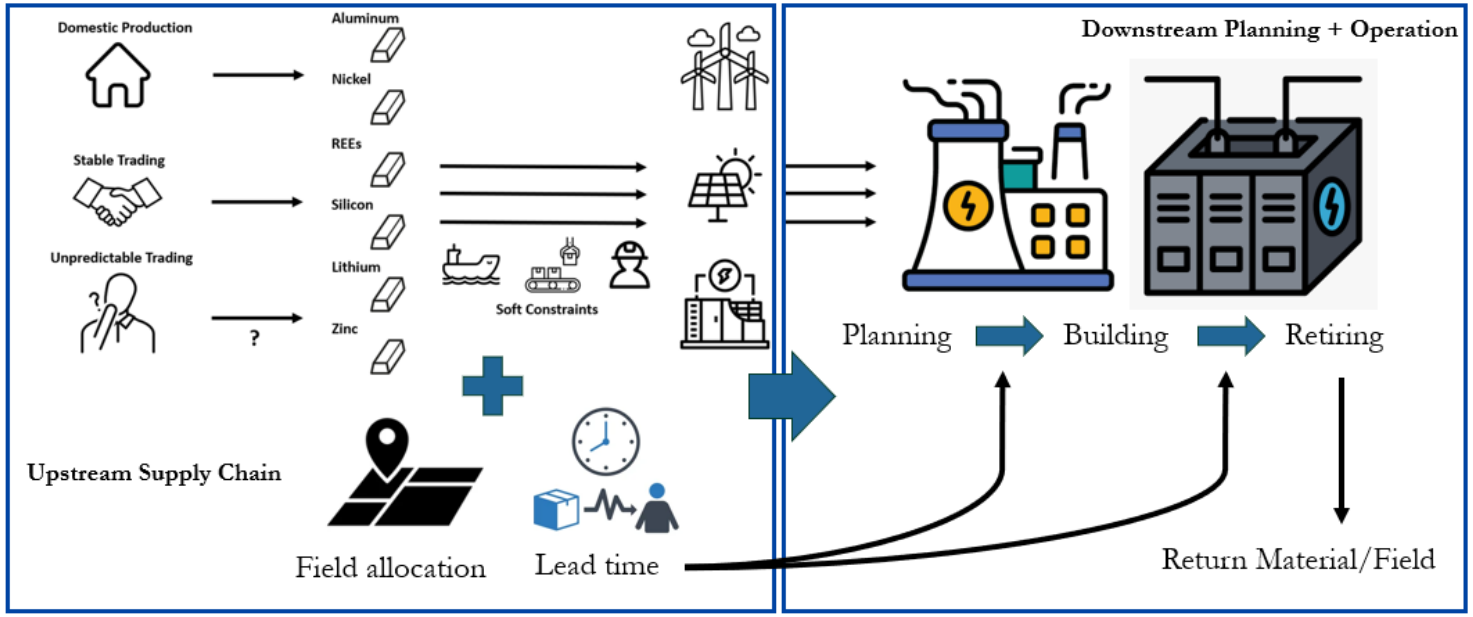
Objective = Investment + Operations + Penalties
Build new capacity, run the system, cover shortfalls.



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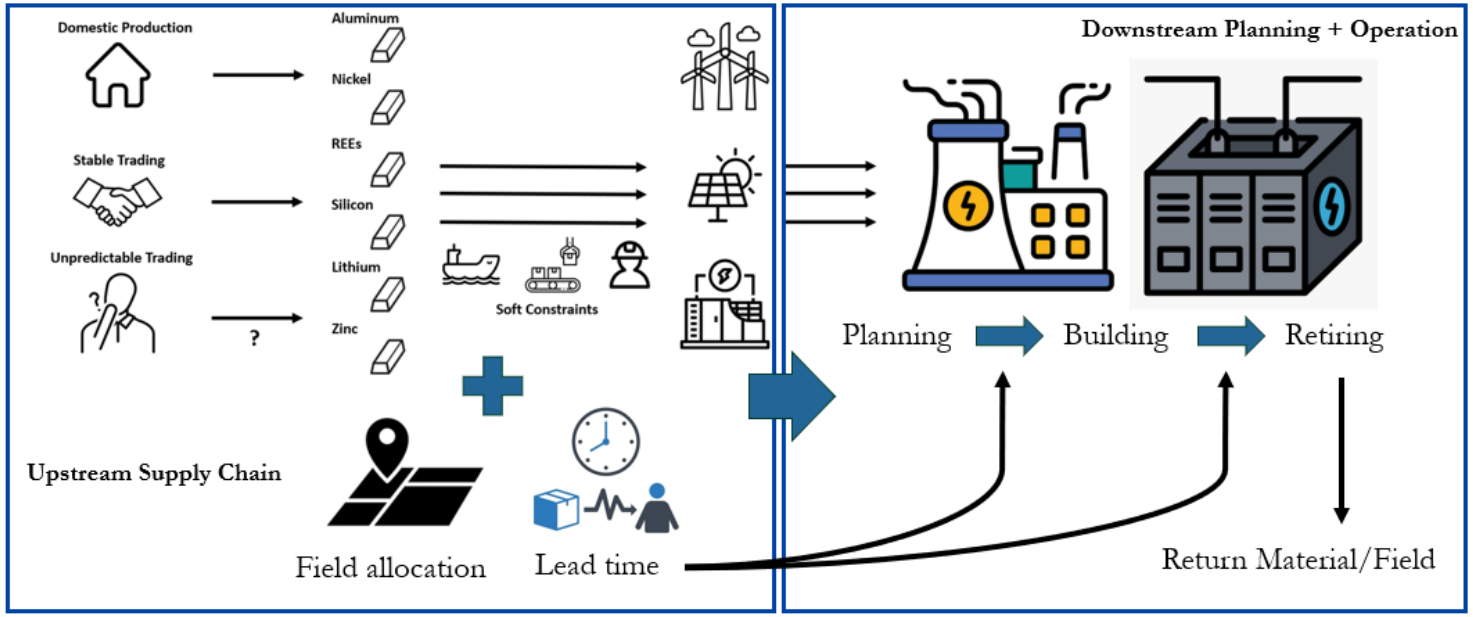

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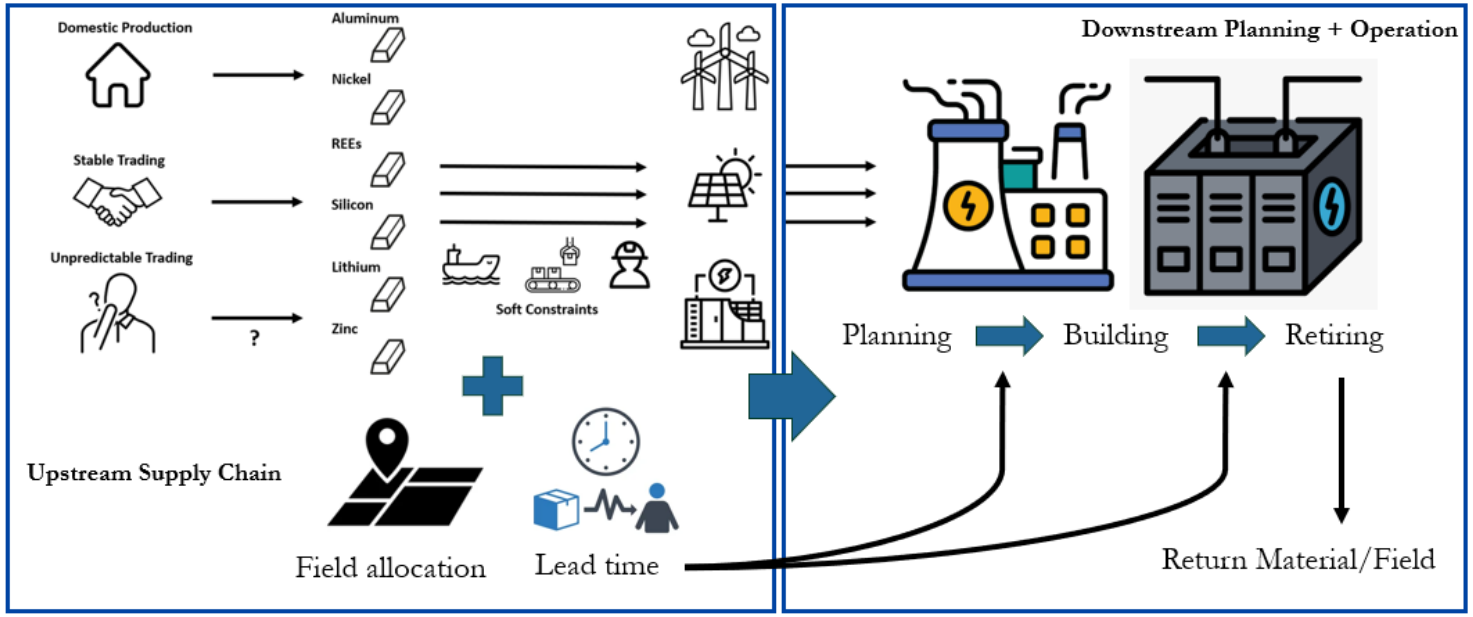

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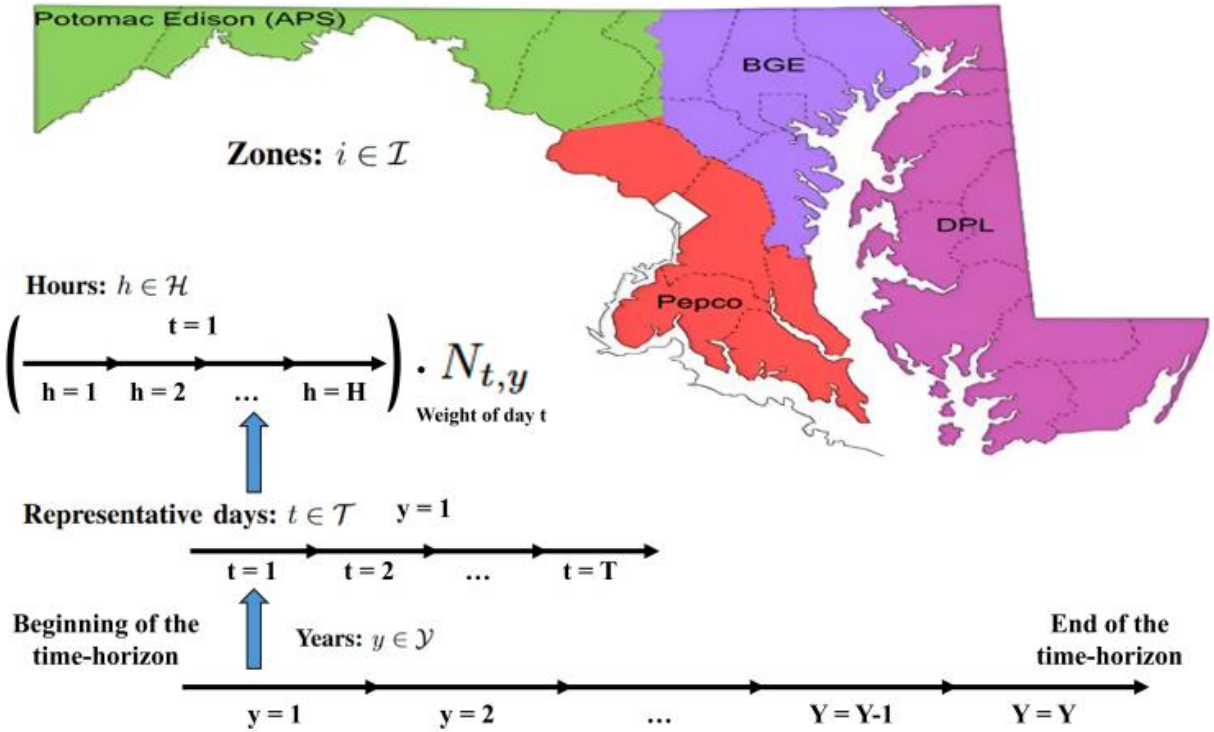
Maryland case study: supply chains shape generation planning



Maryland spatial and temporal representation

- 4 utility zones (BGE, APS, DPL, PEPCO) as nodes.
- Annual investments, hourly ops (rep. days).
- Transmission: 4-node, Window 3 upgrades (fixed)
- Resources: existing + new (renewables + batteries only) (per MD regs & queue).
- Lifetimes: retire at design end.

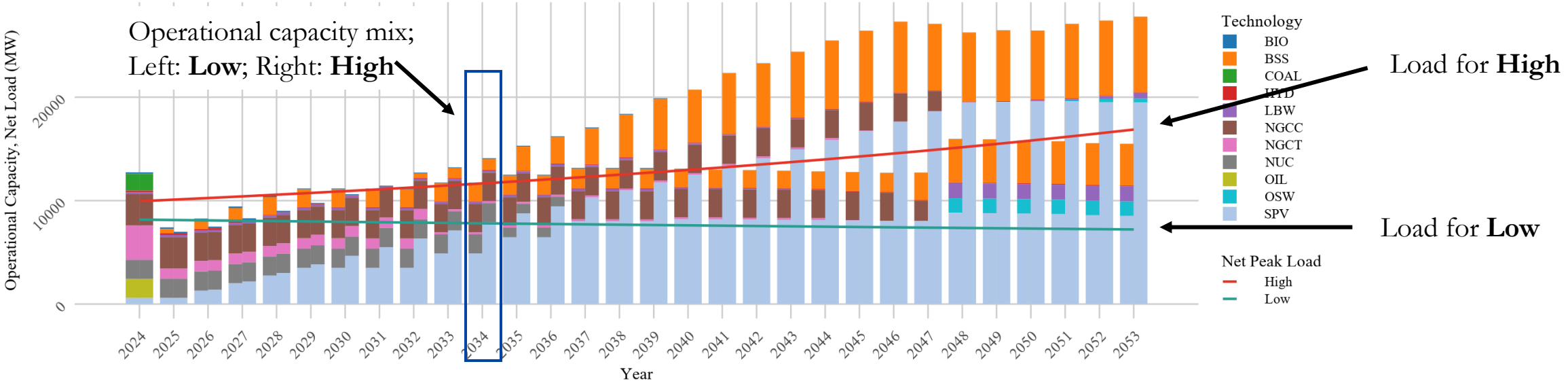
Scenarios:

- **Low**: no data centers, low electrification.
- **High**: with data centers, high electrification.
- Sub-Scenarios:
  - **w/o SC** (relaxed): **unlimited** materials, no lead times.
  - **lim. SC** (constrained): **domestic/allied materials only**, with lead times & land limits.

Replicate traditional generation planning.

Stricter material limits.

Maryland case study: supply chains shape generation planning



Operational capacity over time for baseline Low and High scenarios. Stacked bars: tech-specific capacity (Low baseline = left); Lines: net peak load (system demand)

- Low baseline (2053): 4.1 GW BSS, 1.5 GW LBW, 1.4 GW OSW, 8.5 GW SPV → diversified mix.
- High baseline (2053): 7.3 GW BSS, 0.5 GW LBW, 0.4 GW OSW, 19.5 GW SPV → solar-dominant.

Rising demand + Tighter materials = **Lower flexibility & feasibility**

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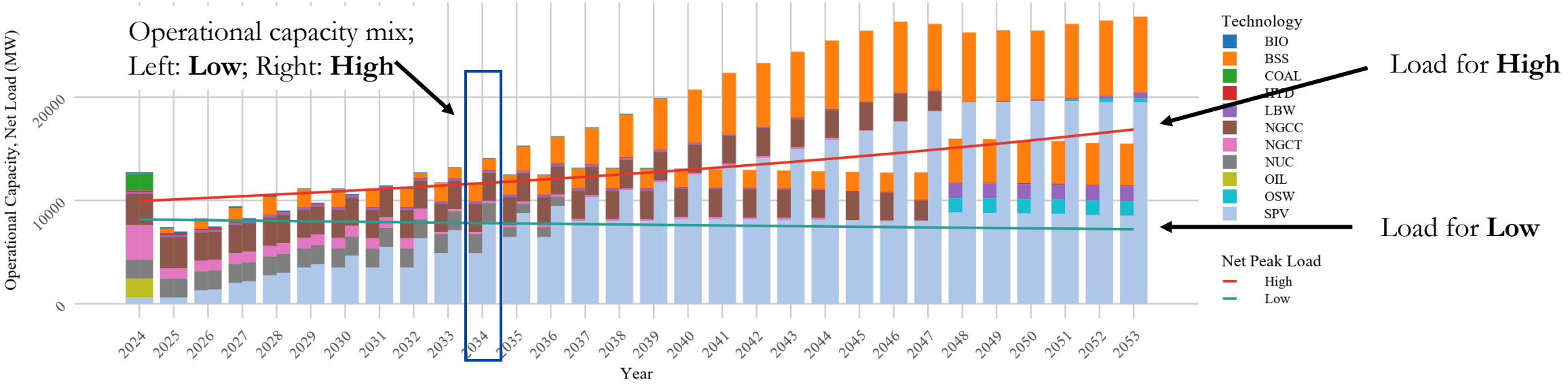

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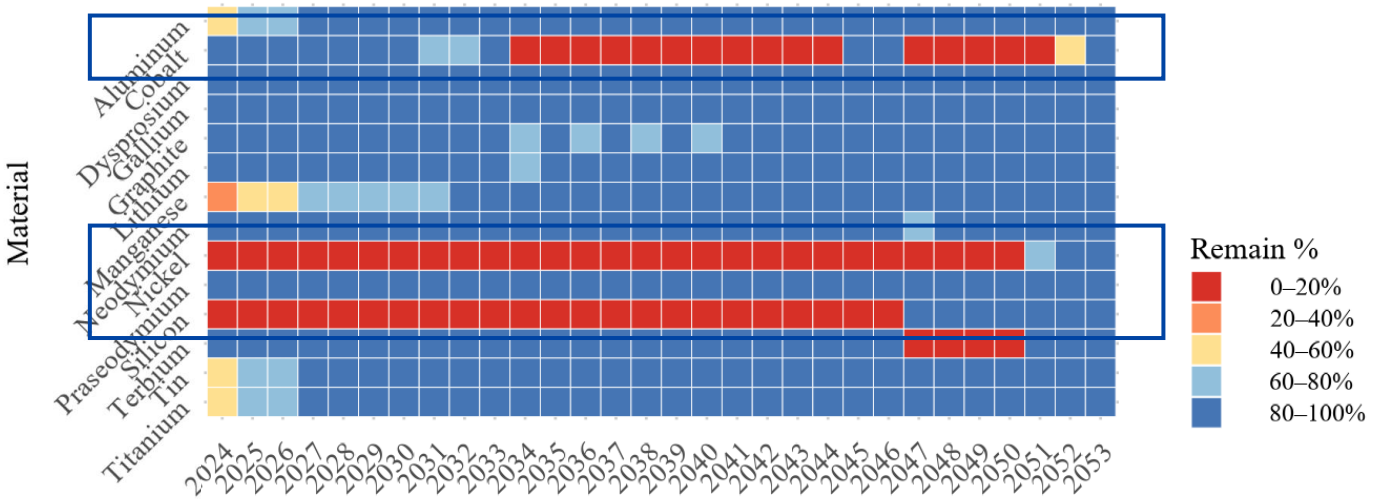
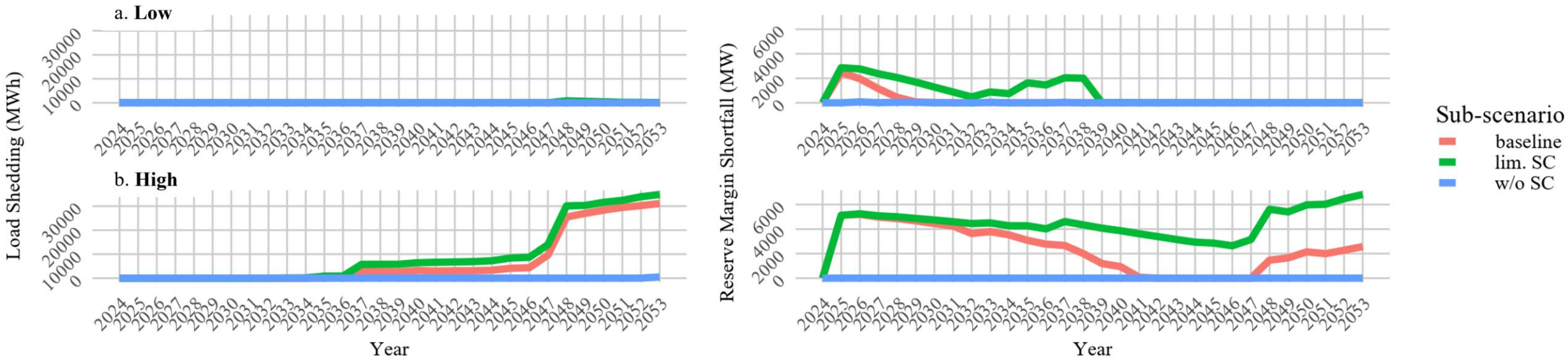
Why this shift?



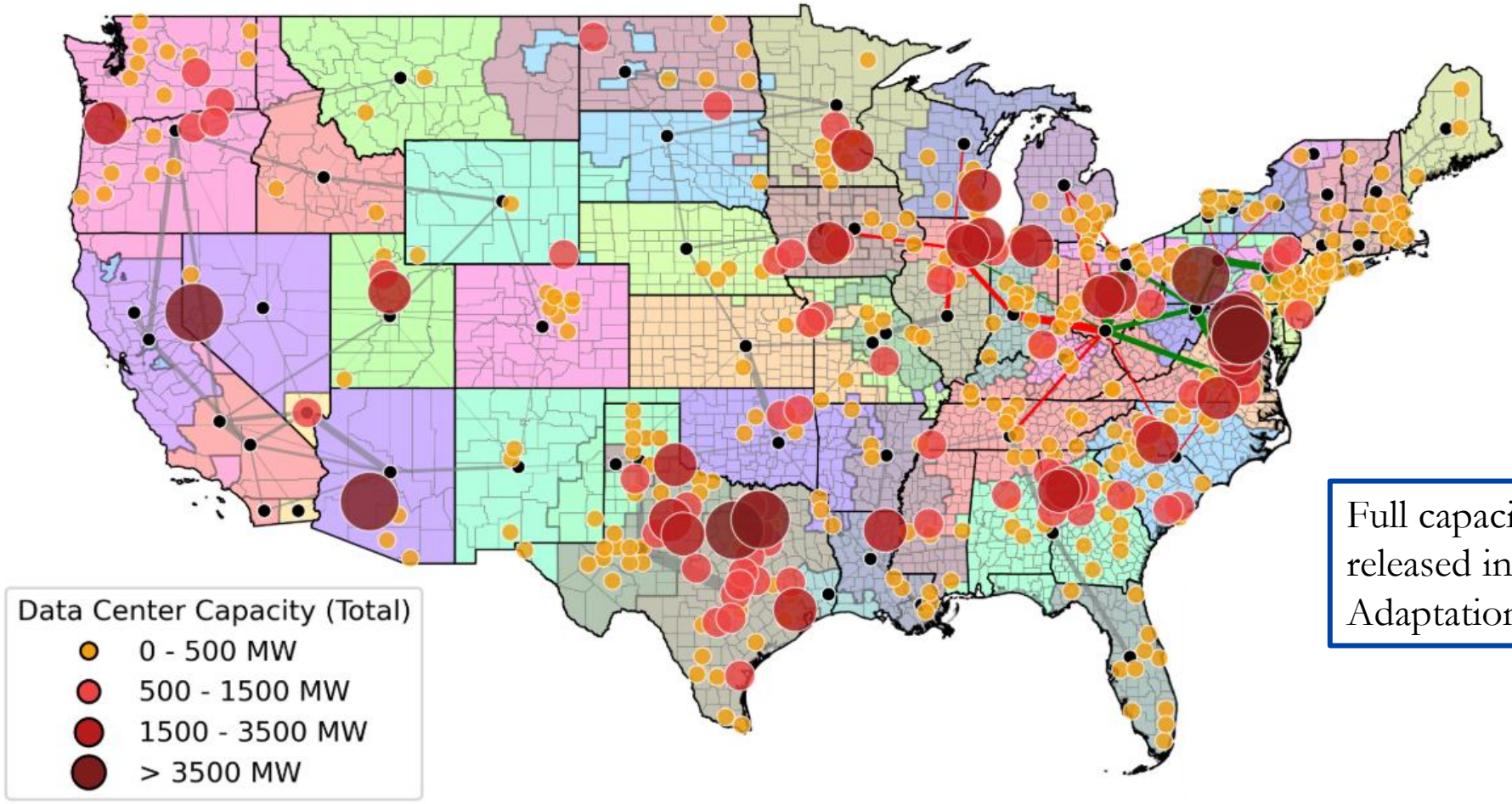
Nickel and cobalt scarcity blocks wind deployment.

Reliability threatened by upstream supply chains



- **Blue lines:** no supply chain limits → just-in-time builds, no physical limits → no load shedding, no reserve margin shortfall.
- **Red lines:** with supply chain limits → insufficient manufacturing capacity → gas plant retirements cause load shedding and reserve margin violations.
- **Green lines:** stricter material limits → fewer resources → greater reliability risk.

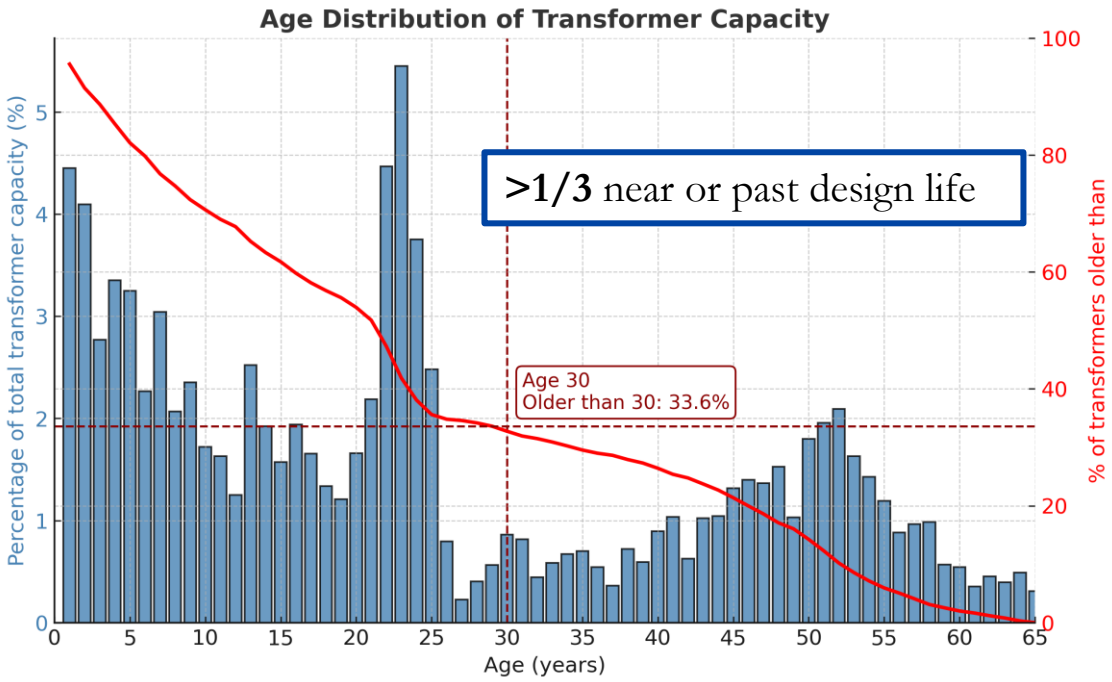
Data center boom strains grid hardware



Full capacity and location data will soon be released in **ICARUS** (Infrastructure Centric Adaptation and Resiliency in the U.S.)!

- U.S. data center capacity → **207.7 GW** (69.8 GW operating, 137.9 GW planned).
- This surge stresses generation, transmission, and **supporting electronics**.

Data center boom strains grid hardware

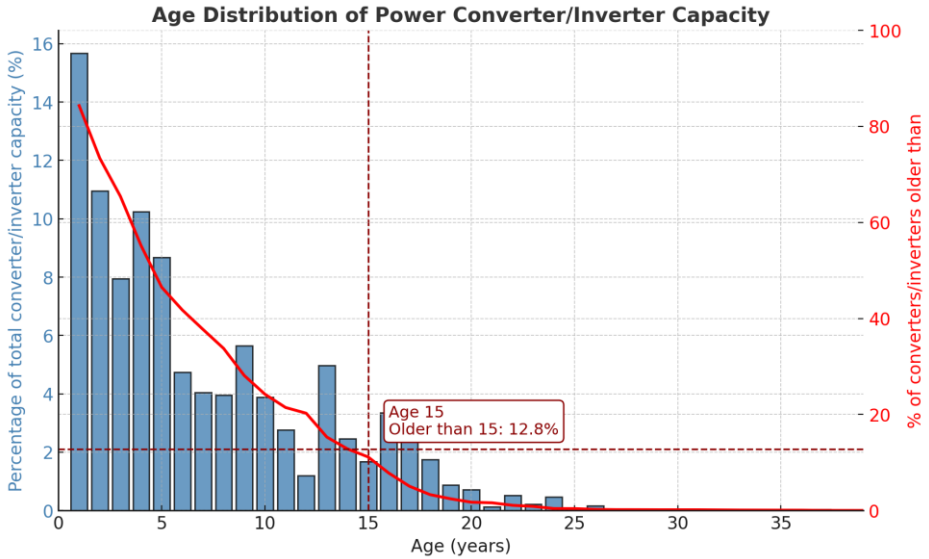


U.S. Large Power Transformer Sources and Market Shares [13]

Source	Unit	2021	2022	2023
U.S. producers	GVA	48.2	53.4	51.3
All import sources	GVA	100.1	124.3	124.5
All sources	**GVA**	**148.3**	**177.7**	**175.8**
U.S. share	%	32.5	30.0	29.2

Annual **demand** (250–300 GVA) far **exceeds supply** (150–180 GVA)

- ~6,746 GVA of transformers in service (transmission, distribution, step-up).
- **~ 1/3 need replacement** or upgrade.
- Imports + domestic **supply ~180 GVA/year, demand > 250 GVA.**
- Converters/inverters are younger, aging wave is coming.
- Supply already lags combined demand for new builds + replacements.



[13] U.S. International Trade Commission. (2024). Large power transformers from South Korea (USITC Pub. 5531). https://www.usitc.gov/publications/701_731/pub5531.pdf

Key takeaways

- Ignoring supply chains underestimates both cost and reliability risks.
- Just-in-time planning is infeasible → requires earlier, strategic investments and lifetime extensions.
- Constraints can erase cost advantages (no materials, no MW) → diversification across technologies is essential.
- Transformers and power electronics are hard limits, not just capital costs in a model.
- Supply chains are emerging as a key determinant of grid and system feasibility.

Thank you!

Yao, B., Jeong, H., Mehrtash, M., Allan, B., Ockerman, D., & Dvorkin, Y. (2025). Understanding supply chain constraints for the US clean energy transition. npj Clean Energy, 1(1), 9.

ICARUS: Infrastructure Centric Adaptation and Resiliency in the U.S.

Yao, B., Bernstein, A., & Dvorkin, Y. (2025). Integrating Upstream Supply Chains into Generation Expansion Planning. *arXiv preprint arXiv:2508.03001.*

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